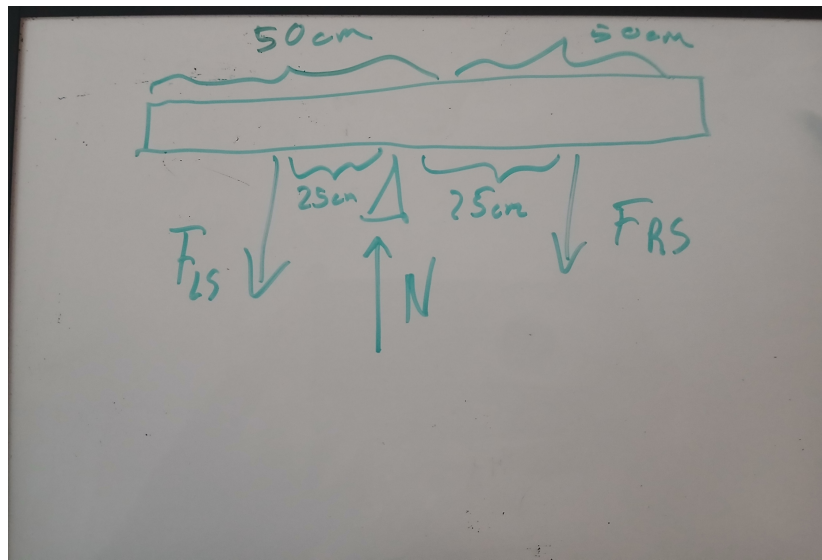


1101W - Asymmetric Balance Warm Up

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1 No mass



1. If we assume that the meter stick is uniformly dense, then the torque per unit length must be uniform. Thus we can simply balance it at its geometric center, halfway along the length of the meter stick.

Using the fact that when determining torque about a fulcrum we can treat a distribution of mass as if centered at the center of mass, we can find the torque of the left side of the meter stick on the entire system. Let's also choose counterclockwise torque to be positive ie. weight pulling down on the left side corresponds to the CCW (counterclockwise) direction.

The forces of weight mg are perpendicular to the meter stick, so the torque from weights here will look like $F \cdot r = mg \cdot r$.

Since the mass of the meter stick is (by assumption) uniformly distributed, the center of mass is merely the geometric center of the left half of the meter stick. Halfway along the left $\frac{1}{2}$ of the length is $\frac{1}{4}$ of the length from

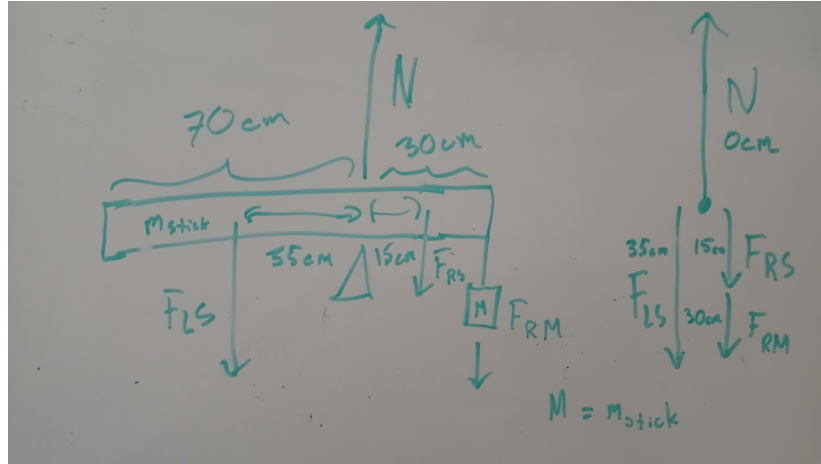
the center. Since half of the total mass is also concentrated in the left half of the meter stick, the resulting torque is

$$\tau_{LS} = m_{LS}g \cdot r_{L,CM} = \left(\frac{1}{2}m_{stick}g\right) \cdot \left(\frac{1}{4}L\right) \quad (1)$$

For the meter stick to not be experiencing angular acceleration (as is necessary in equilibrium) then the torque on the right side of the meter stick must be equal in magnitude but opposite in direction, such that

$$\tau_{RS} = -\tau_{LS} \implies \tau_{NET} \equiv \tau_{RS} + \tau_{LS} = 0 \quad (2)$$

2 One Mass



2. Changing the fulcrum would result in a different torque on the right and the left-hand side. First, let's see what the new torque would look like on the right and left sides

$$\begin{aligned} \tau_{NET} &= \tau_{LS} + \tau_{RS} \\ &= m_{LS}g \cdot r_{L,CM} + (-m_{RS}g \cdot r_{R,CM}) \end{aligned} \quad (3)$$

Note now that the mass on the left and right sides has changed. If (by arbitrary choice) we say that the fulcrum is now 20 cm further to the right side, then the left side is 70 cm long and the right side is 30 cm long. Thus, since the density of the meter stick is uniform, $m_{LS}g = \frac{70cm}{100cm}m_{stick}g = 0.7 \cdot m_{stick}g$. Similarly, $m_{RS}g = \frac{30cm}{100cm}m_{stick}g = 0.3 \cdot m_{stick}g$. From there we find that the center of mass for the left side is $\frac{70cm}{2} = 35cm = 0.35L$ from the center, and the center of mass for the right side is $\frac{30cm}{2} = 15cm =$

$0.15L$. In all we find that

$$\tau_{NET} = 0.7m_{stick}g \cdot (0.35L) - 0.3m_{stick}g \cdot (0.15L) \quad (4)$$

$$= 0.2m_{stick}gL \quad (5)$$

What we need is to add weights in such a way as to add

$$-0.2m_{stick}gL$$

of torque to the system. There are a multitude of possible choices to accomplish this, if you have freedom to choose the mass you attach. One option is to add a mass of the same mass as the meterstick to the right side $20cm$ from the fulcrum, as the resulting torque is

$$-m_{stick}g \cdot 20cm = -m_{stick}g \cdot 0.2L$$

where the $-$ indicates torque in the clockwise direction. We could also add a mass of $\frac{2}{3}m_{stick}g$ to the right side, at a distance from the fulcrum of $30cm$ (all the way at the end). This gives a torque of

$$-0.3L \cdot \left(\frac{2}{3}m_{stick}g\right) = -0.2m_{stick}gL$$

which will cancel the net torque to 0.

3. See the diagram from 2
4. Generalizing the equations above (1)

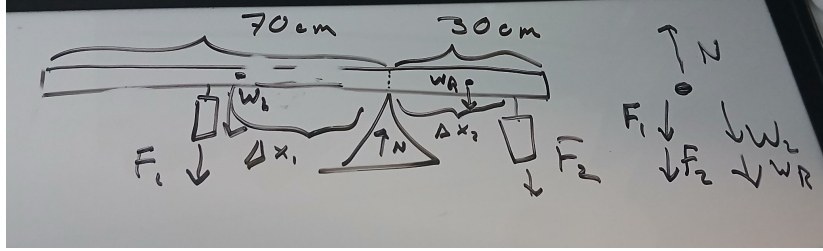
$$\begin{aligned} \tau_{NET} &= \tau_{LS} + \tau_{RS} = 0 \\ &= m_{LS}gr_{LS} - m_{RS}gr_{RS} - m_wgr_w = 0 \end{aligned} \quad (6)$$

5. As before, we use the fact that for a length of the meterstick the mass associated with that extent is $\frac{L_S}{L}m_{stick}$, and its center of mass is halfway along that extent.

$$\begin{aligned} &= m_{stick} \frac{L_{LS}}{L} g \frac{L_{LS}}{2} - m_{stick} \frac{L_{RS}}{L} g \frac{L_{RS}}{2} - m_wgr_w = 0 \\ &= \frac{m_{stick}g}{2L} (L_{LS}^2 - L_{RS}^2) - m_wgr_w = 0 \end{aligned} \quad (7)$$

The net forces are trivial, because we assume that the balance as a whole isn't moving so we just assume all of the weights are countered by the normal force. Solving (7) for r_w

$$r_w = \frac{m_{stick}}{2Lm_w} (L_{LS}^2 - L_{RS}^2) \quad (8)$$



3 Two masses

The net torque equation is effectively the same as above, with one added mass

$$\begin{aligned}
 \tau_{NET} &= \tau_{LS} + \tau_{RS} = 0 \\
 &= F_1 \Delta x_1 + W_L \frac{L_{LS}}{2} - F_2 \Delta x_2 - W_R \frac{L_{RS}}{2} = 0 \\
 &= m_1 g \Delta x_1 + m_{stick} \frac{L_{LS}}{L} \frac{L_{LS}}{2} - m_2 g \Delta x_2 - m_{stick} \frac{L_{LS}}{L} \frac{L_{RS}}{2} = 0
 \end{aligned}$$

We can divide out g from each term, then simplify

$$= m_1 \Delta x_1 - m_2 \Delta x_2 + m_{stick} \frac{1}{2L} (L_{LS}^2 - L_{RS}^2) = 0 \quad (9)$$

4 Prediction

There is a whole space of possible distances one can get for a given pair of masses. We can solve for Δx_1 with respect to Δx_2 and all of the other known quantities from (9).

$$\Delta x_1 = \frac{m_2}{m_1} \Delta x_2 - \frac{m_{stick}}{m_1} \frac{1}{2L} (L_{LS}^2 - L_{RS}^2) \quad (10)$$